

and video streaming of movies are some other Internet applications that are in great demand.

Advertising through the mobile Web is another growth area. There is a lot of demand for value-added content but the need is of a capable network that can deliver content and services.

Features that matter

Experiencing the Internet on a mobile phone is different from the real Internet experience. A mobile phone is an extremely complex device with limitations of screen size and keypad. To actually enjoy browsing, you need a decent screen size on a mobile phone. "A 5.6cm liquid crystal display (LCD) is fairly suitable for the mobile Internet. If the screen size is smaller, it becomes cumbersome to scroll and view content," says Jain. A bigger size is definitely better but it adds to the cost.

"A touchscreen phone helps in better navigation in the browser," claims Prem Kumar, CEO, Fly. "The quality of display lens is important. A capacitive touchscreen provides multi-touch functionality as well. For someone who uses the Internet a lot on a mobile phone, a multi-touchscreen display would be preferred. This makes surfing the Web easier and prevents struggling with small keypads while clicking," adds Kumar.

Most of the websites use client-side scripting and the storage of cookies for an improved user experience. Java, Flash and other third party solutions need to be designed for mobile phone operating systems to support the modern Web. Java-enabled phones are what you should look for. Flash lite is another software that is forging ahead to run perfectly on mobile Internet devices. This is required to view videos on mobile phones.

Another important factor that controls your access to the Internet on a mobile phone is the mobile Web browser. The browsers greatly differ in features and the operating systems they are supported with. An ideal



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browser should have features like multi-tabs, zoom in and compressed downloads for fast browsing, bookmarking and customisation. Opera Mobile offers the multi-tab feature while Opera Mini offers compression for fast browsing. Google Android and Safari provide a rich display of websites. Mozilla's Minimo and Microsoft IE for mobiles are some other popular mobile Web browsers.

Mobile Internet technologies

The 2G world. Presently, most people in India either use general packet radio service (GPRS) or EDGE to access the Internet on a mobile phone. GPRS is a 2G service of the GSM network. The speed is around 25 to 40 kbps (it can ideally support up to 115 kbps), whereas the enhanced data rate for the GSM network (EDGE) can support speeds of 256 kbps. The data rates vary depending on the network strength, traffic and the capability of the mobile handsets.

The 3G world. "If you have a 3G handset, and you use it with the

existing Airtel or Vodafone services that still operate in 2G, you will still get the speed of EDGE, which is 256 kbps," says Jain. With the 3G mobile phone, you need a 3G network to allow you to enjoy high-speed Internet applications. 3G networks will enable operators to offer a wider range of services such as video calls and live TV, which are not yet available in the Indian market.

There are different technologies in the GSM and CDMA standards that work in the 3G spectrum to offer different uplink and downlink data rates. The wide-band code division multiple access (WCDMA) of GSM provides maximum downlink speeds of 384 kbps. High-speed downlink packet access (HSDPA) extends and improves on WCDMA, taking the speed to 3.6 Mbps.

Most early 3G handsets marketed in India were WCDMA. Nokia N70 and N73, Motorazr V3XX, LG Ku990 are some of the 3G phones that support WCDMA and not HSDPA. All the latest varieties of 3G phones like Apple's iPhone, Nokia N96/N97/E71,